



Vandan Nagori

Roll No.:230021020

B.Tech

Space Science and Engineering

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EDUCATION

Degree/Certificate	Institute/Board	CGPA/Percentage	Year
B.Tech. SSE	Indian Institute of Technology Indore	8.58 (Current)	2023-Present
Senior Secondary	CBSE Board	93.2%	2022
Secondary	CBSE Board	86.2%	2020

EXPERIENCE

- **PRIUS Fellowship** *Office Memorandum*
 - Developed Python-based data processing pipeline to analyze galaxy properties from COSMOS survey datasets.
 - Implemented EAZY-py library for photometric redshift estimation using SED template-fitting algorithms.
 - Performed data preprocessing, parameter tuning, and generated UVJ diagrams for galaxy classification analysis.
- **ISRO-NRSC Challenge** *Output, Certificate*
 - Built deep learning segmentation model using UNet architecture for cloud and shadow detection in satellite imagery.
 - Developed pipeline including TOA reflectance conversion, data masking, and normalization techniques.
 - Training workflow with batch processing, loss optimization, and model evaluation metrics implementation.

PROJECTS

- **Immersive 3D Portfolio (Space-Themed)** *Link*
 - Developed a space-themed interactive 3D portfolio using React Three Fiber and Three.js.
 - Optimized 3D performance with Draco compression, lazy loading, and code splitting.
 - Enhanced visuals using custom GLSL shaders, HDR lighting, and procedural starfields.
 - Engineered reusable 3D components with smooth transitions, camera interpolation, and docking animations.
- **WanderLust (Airbnb Clone)** *Link, GitHub*
 - Built a full-stack platform Wanderlust using Node.js, Express, MongoDB, and EJS with CRUD operations.
 - Designed RESTful APIs using MVC architecture with validation and Cloudinary-based image handling.
 - Implemented secure authentication and role-based access using Passport.js and session management.
 - Deployed on Vercel with MongoDB Atlas and managed CI/CD using Git and GitHub.
- **Club Web Platforms (IIT Indore)** *Gaming, Astronomy*
 - Led web development team to build and deploy Gaming Club and Astronomy Club websites using React.js.
 - Coordinated with team members for task distribution, code reviews, and collaborative development using Git.
 - Developed responsive UI components with ReactJs and TailwindCSS ensuring seamless cross-device compatibility.
 - Managed deployment pipeline on Vercel and maintained consistent uptime through regular monitoring and updates.
- **Academic Management System** *GitHub*
 - Developed a full-stack academic system using Node.js, Express, MongoDB, and EJS with role-based dashboards.
 - Implemented student features including timetable management, quizzes, grades, and homework tracking.
 - Built faculty tools for content management, quiz creation, assignments, and student resource control.

TECHNICAL SKILLS

- **Programming:** Python, C/C++, *DSA
- **Web Dev:** HTML, CSS, Javascript, ReactJS, NodeJS, Express, EJS, Bootstrap, Tailwind CSS, Three.JS*
- **Databases:** MongoDB, SQL
- **Tools:** Git/GitHub, Terminal ** Elementary proficiency*

KEY COURSES TAKEN

- **Mathematics:** Linear Algebra, Differential Equations, Complex Analysis, Numerical Methods and Calculus
- **Core:** Digital Electronics, Analog Electronics, Electronics Lab, Scientific Computing (python)
- **Development:** Web Development (Apna College) *Certificate*

POSITIONS OF RESPONSIBILITY

- **Head of Web Development,** Astronomy Club, IIT Indore *Jul. 2023 - Jul. 2025*
- **Head of Technicals,** Gaming Club, IIT Indore *Dec. 2024 - Apr. 2025*